Data Warehouse Design - Justification

Task 3

Group 20

1. Fact Tables and Grains

Fact table	Grain	Measures / subject
FACT_ECONOMY	Country × Year	GDP, GDP per capita, growth and trade shares
FACT_SOCIETY	Country × Year	Population, GNI per capita, life expectancy and urbanization
FACT_SECTOR	Country × Year × Sector	Value added (% GDP and USD) and employment share

Economy and society have the same country–year grain but different subjects. Sector data has a finer grain; merging the facts would repeat country-level measures across sectors and risk double-counting.

2. Dimensions and Keys

DIM_COUNTRY and **DIM_TIME** are conformed dimensions shared by the three facts, allowing consistent country/group and year/decade analysis. All dimensions use surrogate primary keys; ISO3 code, year and sector name remain unique business identifiers.

DIM_SECTOR connects only to **FACT_SECTOR**, because national economy and society measures have no sector-level detail. The hierarchy is Primary: Agriculture; Secondary: Industry (Manufacturing included); Tertiary: Services. Manufacturing is part of Industry and must not be added to it.

3. Historical Events Integration

The assignment allows three ways to integrate historical events:

Approach	Why choose it (or not)?
Reference table joined by year	Simplest choice, but does not allow easy searches for events by country.
Country–time bridge (chosen)	Connects events to country and year, enabling both specific and broader historical searches; supports multiple events per country–year.
DIM_EVENT dimension	Organizes event attributes, but needs additional links to associate events with country–year observations.

Choice: This project uses historical events primarily to help explain economic changes. Events therefore need to be linked to both country and year, so we can examine changes over time in general or for a specific country. **BRIDGE_EVENT** stores each event separately, with foreign keys to **DIM_COUNTRY** and **DIM_TIME**, as well as its description, category and economic impact.

Aggregation safeguard: Multiple events may share one country–year, so a direct join with a fact table could repeat GDP or other measures. Retrieve events separately or join only after aggregating the facts.